

#14Days14Chapters

DAY 7

[Aim: 100/100 in Maths]

Class 10th → Mean, Median, Mode
(Average)

STATISTICS

Types of Data

Ungrouped Data / Raw data

A → 4

B → 5

M → 6

G → 8

Grouped data

Marks	No. of students
0 - 10	4
10 - 20	10
20 - 30	5
30 - 40	2
40 - 50	1

#Ungrouped Data:

#Mean:

$$\text{Mean/Average} = \frac{\text{Sum of all observations}}{\text{No. of observ.}}$$

($\bar{X}$)

#Ex:- Marks of 8 students:-

4, 2, 8, 6, 5, 7, 2, 1

$$\text{Mean} = \frac{4+2+8+6+5+7+2+1}{8}$$

$$\bar{X} = \frac{35}{8}$$

#LP:

If the arithmetic mean of the first n natural numbers is 15, then n is _____.

$$\bar{x} = 15$$

[CBSE 2020]

$$\Rightarrow \frac{\text{sum of all 'n' natural no.}}{\text{no. of observations}} = 15$$

$$\Rightarrow \frac{1+2+3+4+5-\dots+n}{n} = 15$$

$$\Rightarrow \frac{\cancel{n} \frac{1}{2} (2(1) + (n-1)(1))}{\cancel{n}} = 15$$

$$\Rightarrow \frac{1}{2} (2+n-1) = 15$$

$$1+n = 15 \times 2$$

$$1+n = 30$$

$$\boxed{n = 29} //$$

#Ungrouped Data:

#Median:

→ middle value but first Ascending/Descending order

#Ex: Marks of 9 students Find median:-

~~2~~, ~~1~~, ~~3~~, ~~7~~, ~~6~~, ~~2~~, ~~8~~, ~~3~~, ~~5~~

Ascending:- 2, 2, 3, 3, 4, 5, 6, 7, 8

middle value

∴ Median = 4

#1P: Marks of 10 students, find median:-

~~2~~, ~~7~~, ~~6~~, ~~9~~, ~~4~~, ~~3~~, ~~1~~, ~~0~~, ~~7~~, ~~8~~

Ascending $\Rightarrow$ 0, 1, 2, 3, 4, 6, 7, 7, 8, 9

$$\text{Median} = \frac{4+6}{2} \Rightarrow \textcircled{5}$$

#Ungrouped Data:

#Mode: (जिसकी freq. सबसे ज्यादा है)

#Q: Marks of 10 students, find mode:-

2, 3, 4, 2, 2, 5, 6, 7, 3, 2

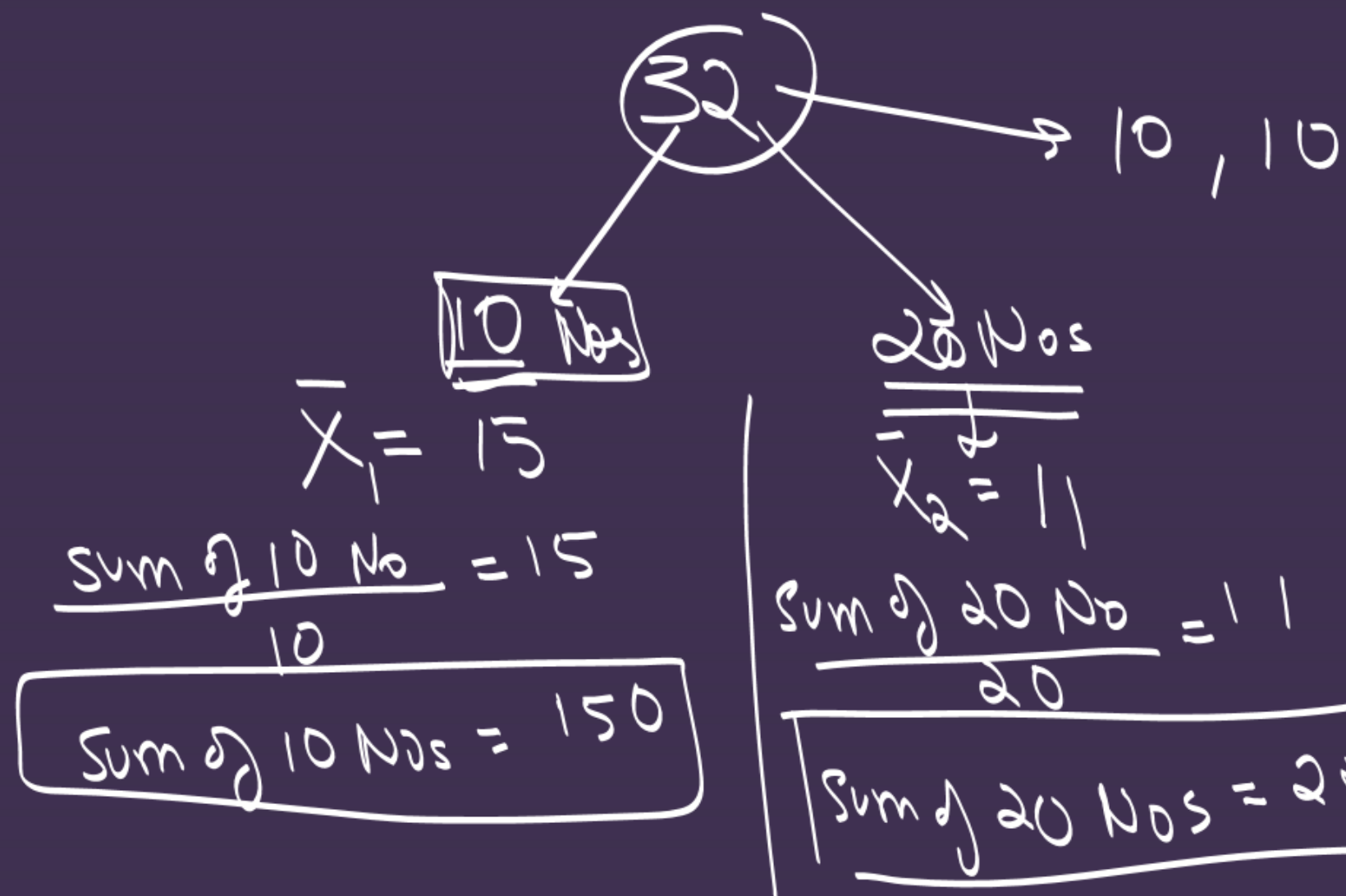
	freq
3	2
2	4

Mode ≥ 2

#LP: Find the mean of the 32 numbers, such that if the mean of 10 of them is 15 and the mean of 20 of them is 11. The last two numbers are 10.

Final
Ans

$$\text{Mean of 32 numbers} = \frac{\text{Sum of 32 numbers}}{32}$$
$$(\bar{X})$$



32 no. sum

$$\text{Mean of 32 No} = \frac{150 + 220 + 10 + 10}{32}$$

$$\bar{X} = \frac{390}{32} = 12.1875$$

$$\bar{X} = 12.1875$$

#Grouped Data:

Mean ✓

Median ✓

Mode ✓

Exclusive

Class
Interval (CI) | ~~Upper Limit (UL)~~
freq. (f)

Lower
Limit

0 - 10

2

10 - 20

3

20 - 30

7

30 - 40

8

40 - 50

9

Class
Size (h)

$\Rightarrow UL - LL$

$(LL - 0.5)$
 $(UL + 0.5)$

Exclusive

0.5 - 10.5

CI

1 - 10

2

10.5 - 20.5

11 - 20

3

20.5 - 30.5

21 - 30

4

30.5 - 40.5

31 - 40

9

40.5 - 50.5

41 - 50

8

Inclusive

(Not in syllabus)

#Mean of Grouped Data:

Direct Method

① $x_i^o = \frac{UL + LL}{2}$

② $f_i^o x_i^o$

③ $\bar{X} = \frac{\sum f_i x_i}{\sum f_i}$

Assumed Mean Method

① x_i^o

② $d_i^o = x_i^o - a$

③ $f_i^o d_i^o$

④ $\bar{X} = a + \frac{\sum f_i d_i^o}{\sum f_i}$

Step-deviation method

① x_i^o

② d_i^o

③ $U_i^o = \frac{d_i^o}{h}$

④ $f_i^o U_i^o$

⑤ $\bar{X} = a + \left(\frac{\sum f_i U_i^o}{\sum f_i} \right) h$

#LP: Find the mean using direct method.

Class	Frequency (f_i)	$X_i = \frac{UL + LL}{2}$	$f_i X_i$
0-10	12	$\frac{0+10}{2} \Rightarrow (5)$	60
10-20	18	$\frac{10+20}{2} \Rightarrow (15)$	270
20-30	27	$\frac{30+20}{2} \Rightarrow (25)$	675
30-40	20	$\frac{30+40}{2} \Rightarrow (35)$	700
40-50	17	45	765
50-60	6	55	330
	<u>100</u>		<u>2800</u>

$$\bar{X} = \frac{\sum f_i X_i}{\sum f_i}$$

$$\bar{X} = \frac{2800}{100}$$

$$\boxed{\bar{X} = 28}$$

#LP: Find the mean using assumed mean method.

Class	Frequency	x_i	$d_i = x_i - a$	$f_i d_i$
0-10	12	5	$5 - 25 = -20$	-240
10-20	18	15	$15 - 25 = -10$	-180
20-30	27	25	$25 - 25 = 0$	0
30-40	20	35	10	200
40-50	17	45	20	340
50-60	6	55	30	180
	100			300

$$\bar{x} = a + \frac{\sum f_i d_i}{\sum f_i}$$

$$\bar{x} = 25 + \frac{300}{100}$$

$$\bar{x} = \frac{2500 + 300}{100}$$

$$\bar{x} = 2800$$

$$\bar{x} = 28$$

#LP: Find the mean using step deviation method.

Class	Frequency	x_i^0	$d_i^0 = x_i^0 - 25$	$U_i^0 = \frac{d_i^0}{h}$	$f_i U_i^0$
0-10	12	5	-20	$\frac{-20}{10} = -2$	-24
10-20	18	15	-10	$\frac{-10}{10} = -1$	-18
20-30	27	25 ← a	0	0	0
30-40	20	35	10	1	20
40-50	17	45	20	2	34
50-60	6	55	30	3	18
					30

$$h = UL - LL = \frac{100}{10} = 10 = h$$

$$\bar{X} = a + \left(\frac{\sum f_i U_i^0}{\sum f_i} \right) h$$

$$\bar{X} = 25 + \frac{30}{100} \times 10$$

$$\bar{X} = 28$$

#LP: Find the value of p , if the mean of the following distribution is 7.5. (CBSE 2020)

Classes	Frequency
2-4	6
4-6	8
6-8	15
8-10	p
10-12	8
12-14	4

(f_i)

x_i

$f_i x_i$

3

~~18~~

5

~~40~~

7

105

9

9p

11

~~88~~

13

52

$$\bar{x} = 7.5$$

$$\frac{\sum f_i x_i}{\sum f_i} = 7.5$$

$$\frac{303 + 9p}{41 + p} = 7.5$$

$$p = 3$$

$$41 + p$$

$$303 + 9p$$

#LP: Find the values of x and y if the mean and total frequency of the distribution are 25 and 50 respectively.

Class Interval	Frequency (f_i)	x_i^o	$d_i^o = x_i - a$	$U_i^o = \frac{d_i^o}{h} = \frac{d_i^o}{10}$	$f_i U_i^o$
0 - 10	7	5	-20	-2	-14
10 - 20	<u>x</u>	15	-10	-1	-x
20 - 30	5	<u>25</u> $\leftarrow a$	0	0	0
30 - 40	y	35	10	1	y
40 - 50	4	45	20	2	8
50 - 60	2	55	30	3	6
<u>$\sum f_i = 50$</u>					

$$\bar{X} = 25$$

$$\bar{X} = a + \left(\frac{\sum f_i U_i}{\sum f_i} \right) h$$

$$25 = 25 + \left(\frac{y - x}{50} \right) \times 10$$

$$0 = \frac{y - x}{5}$$

$$0 = y - x$$

$$\boxed{y = x}$$

$$\sum f_i = 50$$

$$y = 18 \quad 18 + x + y = 50$$

$$x = 16 \quad x + x = 32 \quad 2x = 32 \quad x = 16$$

#1P:

Find the mean of the following data:

[CBSE 2013]

Inclusive

Area (in hectares)	No. of farms <i>(f_i)</i>
1-3	25
4-6	22
7-9	52
10-12	45
13-15	16

(8)

x_i	$d_i = x_i - 8$	$f_i d_i$
2	-6	-150
5	-3	-66
<u>8</u> $\leftarrow a$	0	0
11	3	135
14	6	96

$$\bar{X} = a + \frac{\sum f_i d_i}{\sum f_i}$$

#LP: Find the mean marks:

Class Interval	Number of students	CI	f	x_i
0 and above	80	0-10	3	
10 and above	77	10-20	5	
20 and above	72	20-30	7	
30 and above	65	30-40	10	
40 and above	55	40-50	12	
50 and above	43	50-60	15	
60 and above	28	60-70	12	
70 and above	16	70-80	6	
80 and above	10	80-90	2	
90 and above	8	90-100	8	
100 and above	0			

$$\bar{X} = \frac{\sum f_i x_i}{\sum f_i}$$

#Median: of Grouped data

$$\text{Median} = l + \left(\frac{\frac{N}{2} - cf}{f} \right) h$$

l = L.L of median class

N = No. of Obsv. (f)

f = freq. of median class

cumulative
freq.

cf = c.f. of class preceding median class (*)

h = class size

#LP: Calculate median marks of the following data:

$$\text{Median} = l + \left(\frac{\frac{N}{2} - cf}{f} \right) h$$

Classes	Number of students ^(f)	cf
0 - 10	2	2
10 - 20	12	14
<u>20 - 30</u>	<u>22</u>	<u>36</u>
30 - 40	8	44
40 - 50	6	50

Median class

$$\begin{aligned} l &= 20 \\ N &= 50 \\ f &= 22 \\ cf &= 14 \\ h &= 10 \end{aligned}$$

How to find median class?

1. cf
2. $\frac{N}{2}$
3. $\frac{N}{2}$ just $\frac{N}{2}$ → median class

$$\frac{N}{2} \Rightarrow \frac{50}{2} = 25$$

just $\frac{N}{2}$ cf?

#LP:

If the median of the following data is 32.5, find the missing frequencies.

$$\frac{2N}{2} = \frac{40}{2} = 20 \rightarrow \text{Just after}$$

[CBSE 2019]

$$\text{Median} = 32.5$$

Class Interval	Frequency
0 - 10	f_1
10 - 20	5
20 - 30	9
30 - 40	12
40 - 50	f_2
50 - 60	3
60 - 70	2
Total	$N = 40$

Median class

l

$$\begin{aligned} &cf \\ &f_1 \\ &f_1 + 5 \\ &f_1 + 14 \\ &\cancel{f_1 + 26} \\ &f_1 + f_2 + 26 \\ &f_1 + f_2 + 29 \\ &f_1 + f_2 + 31 \end{aligned}$$

$$f_1 + f_2 + 31 = 40$$

$$f_1 + f_2 = 9 \quad \text{--- (I)}$$

put

$$3 + f_2 = 9$$

$$f_2 = 9 - 3$$

$$f_2 = 6$$

$$\begin{aligned} &l + \left(\frac{\frac{N}{2} - cf}{f} \right) h = 32.5 \\ &30 + \left(\frac{20 - (f_1 + 14)}{12} \right) \times 10 = 32.5 \\ &\Rightarrow \left(\frac{20 - f_1 - 14}{12} \right) \times 10 = 32.5 - 30 \\ &\Rightarrow \left(\frac{6 - f_1}{12} \right) \times 10 = 2.5 \\ &\Rightarrow (6 - f_1) = \frac{2.5}{10} \times 12 \times 3 \\ &\Rightarrow 6 - f_1 = 3 \\ &\Rightarrow f_1 = 3 \end{aligned}$$

$$h = 10$$

#LP:

Find the median of the following data:

Monthly consumption (in units)	No. of Consumers
Below 85	4
Below 105	9
Below 125	22
Below 145	42
Below 165	56
Below 185	64
Below 205	68

[CBSE 2012]

$$\text{Median} = l + \left(\frac{\frac{N}{2} - cf}{f} \right) h$$

$$\Rightarrow 125 + \left(\frac{\frac{68}{2} - 22}{20} \right) 20$$

CI	f	cf
below 85	4	4
85-105	5	9
105-125	13	22
125-145	20	42
145-165	12	56
165-185	8	64
185-205	4	68
N = 68		

Median class

#Mode:

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) h$$

#LP: Find the mode of the following data:

Classes:	0-20	20-40	40-60	60-80	80-100	100-120	120-140
Frequency:	6	8	10	12	6	5	3

L	h	f
0-20		6
20-40		8
40-60		10
60-80		12
80-100		6
100-120		5
120-140		3

Model class

f_0

f_1

f_2

Model class
↓
highest 'f'
class

$$\text{Mode} = L + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) h$$

$$= 60 + \left(\frac{12 - 10}{2(12) - 10 - 6} \right) 20$$

↓

#Relationship b/w Mean, Median & Mode:

$$3 \text{ Median} = 2 \text{ Mean} + \text{Mode}$$

#LP: The mean and median of a statistical data are 21 and 23 respectively. The mode of the data is : (2024)

- ☒ (a) 27
- (b) 22
- (c) 17
- (d) 23

$$\begin{aligned}\text{Mean} &= 21 \\ \text{Median} &= 23 \\ \text{Mode} &= ?\end{aligned}$$

$$3\text{Median} = 2\text{Mean} + \text{Mode}$$

$$3(23) = 2(21) + M$$

$$69 = 42 + M$$

$$69 - 42 = M$$

$$\boxed{M = 27}$$

7/14 Completed!

Thank you Goodies !!

